

AI-Einstein: An AI Framework for Discovering New Scientific Concepts and Theories in Real Historical Scenarios

Abstract. AI-driven scientific discovery has made significant progress nowadays. However, artificial intelligence has not yet been able to discover scientific concepts and theories in new-discovery, and scientific re-discovery research has not effectively demonstrated AI’s capabilities in discovering scientific concepts and theories because it has not utilized data from real-world cases. In this paper, we propose AI-Einstein, a novel AI framework to discover new scientific concepts and theories using historical data in real scientific discovery scenarios where multiple hypotheses coexist. The framework is divided into three parts: the first part, inspired by the philosophy and history of science, expands the theoretical space through ‘take back’ operation; the second part utilizes an auto-encoder architecture to ascertain how the predicted values of scientific hypotheses should be adjusted to align with actual observational data; the third part employs external constraints to filter and identify the most optimal scientific hypothesis. To prove our framework works, we apply it to a classic case in the history of physics at the end of the 19th century—the theory of light propagation and measurements regarding ether. Experimental results on real historical data demonstrate that even in scenarios of actual historical discoveries, our AI Framework can suggest the discovery of new scientific concepts and theories, such as the necessity for a quadratic transformation of spatial and temporal parameters and the plausibility of assuming the absence of ether. These findings remarkably align with the fundamental assumptions made by Einstein in his groundbreaking paper on special relativity in 1905.

1. Introduction

Artificial intelligence has been rapidly and widely applied in various fields of scientific research over the past decade[1, 2], bringing new discoveries in data-intensive areas such as astronomy[3], particle physics[4, 5], and structural biology[6]. Generative large models have accelerated this process, not only in various vertical domains [7–9], but general-purpose LLM can also aid in automatic scientific discovery [10, 11], even in the field of mathematics [12]. On the other hand, research on scientific re-discovery has also made progress due to the use of deep learning, especially in the field of physics, where depth can lead to the discovery of scientific concepts[13], formulas[14–17], and even principles[18], including the automation of the entire scientific discovery process [19], as well as studies on the interpretation of scientific thoughts and theories [20].

However, in the field of AI-driven scientific new-discovery, whether it’s traditional machine learning methods or recent generative large models based on the transformer

architecture, there has not yet been a discovery of true new scientific concepts and theories[‡]. Useful new discoveries have been focused on predicting phenomena and extracting patterns from data. In the field of scientific re-discovery, research primarily utilizes data simulated through known physical formulas[14–16] or data obtained under technological conditions far more advanced than those available at the time of the actual discoveries[17, 19]. Models trained with such data cannot demonstrate artificial intelligence’s capabilities in real-world significant scientific discovery scenarios (often characterized by insufficient data and unknown directions for innovation) regarding scientific concepts and theories.

To address these problems, we propose AI-Einstein to understand whether, in real historical scenarios of scientific discovery, machine learning can identify new concepts, theories, and even ideas when data is scarce and not directly related. The framework is divided into three parts. The first part, addressing the coexistence of multiple theories and inspired by the philosophy and history of science, identifies contradictions between competing theories and uses “take back” to relax constraints at these contradiction points, thereby expanding the theoretical space. The second part, based on a classical auto-encoder architecture[21], identifies the relationship between changes in the predictions of different scientific hypotheses and actual observations, determining how different scientific hypotheses need to be modified to align with real-world observations. The third part, using external constraints such as simplicity and symmetry, filters the modification in different scientific hypotheses to find the most suitable hypothesis.

We apply this framework to a classic case—the propagation of light and ether measurements in the 19th century[22]—using real historical data, and find that even in scenarios of real historical discoveries, machine learning methods can at least suggest the discovery of new scientific concepts and theories.

Our contributions are:

- (1) Demonstrating the effectiveness of artificial intelligence in discovering scientific concepts and theories using actual historical data from real cases.
- (2) Proposing a novel AI scientific discovery framework, AI-Einstein, tailored for scenarios where multiple theories coexist, providing a new method for expanding theoretical space and resolving contradictions between theories and observations.
- (3) Applying machine learning to solve a classic problem in the physics revolution of the early 20th century—the ether drift, showcasing the framework’s capability in addressing historic scientific challenges.

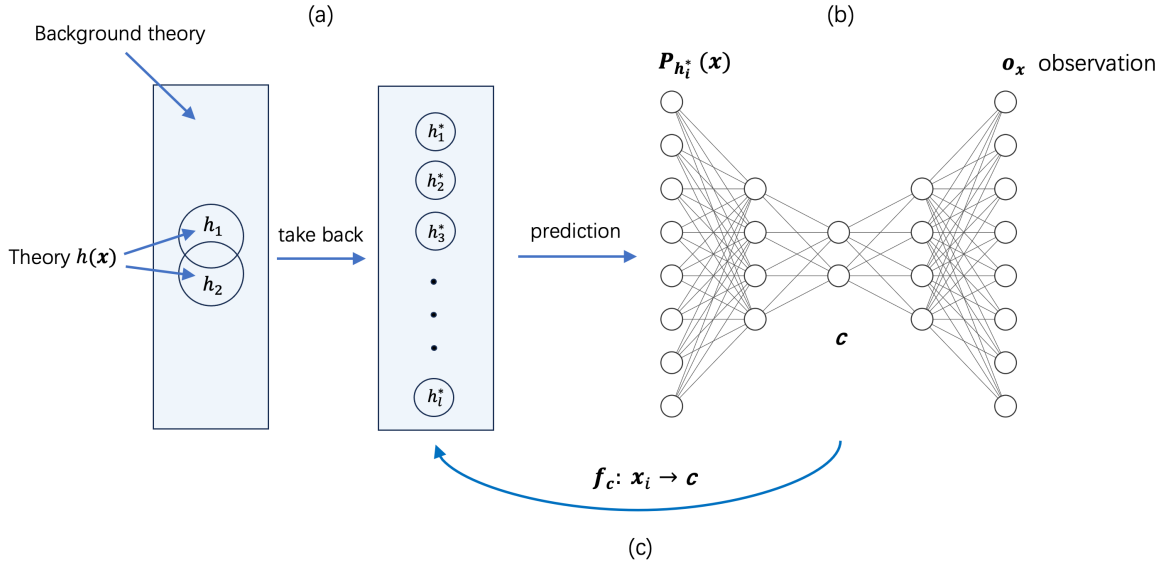


Figure 1: The AI-Einstein Framework for Scientific Discovery in Scenarios with Coexisting Theories. It is consisted with three parts: (a) the ‘take back’ process is used to expand the theoretical space at points where hypotheses contradict each other, (b) the auto-encoder architecture is used to find changes from the predictions of scientific hypotheses to real data, (c) identify the most optimal hypothesis by comparing how different scientific hypotheses should change.

2. Methods

2.1. The AI-Einstein Framework

It consists of three main components: expanding the theoretical space through the ‘take back’ operation, utilizing an auto-encoder architecture to determine how predictions of hypotheses should be adjusted to align with observational data, and employing external constraints to identify the most optimal hypothesis.

2.1.1. Theoretical Space Expansion by ‘Take Back’ Operation The first part of the AI-Einstein Framework focuses on expanding the theoretical space where multiple competing theories coexist. We represent the scientific hypothesis used to explain physical phenomena with $h_i(\mathbf{x})$, $\mathbf{x} = (x_1, x_2, \dots, x_n)$, where $\mathbf{x}$ represents the relevant variables of the phenomenon, such as time t , spatial distance l , etc. The predicted values of the variables given by the hypothesis are $\mathbf{p}_{h_i}(\mathbf{x}) = (p_{h_i}(x_1), p_{h_i}(x_2), \dots, p_{h_i}(x_n))$, and the actual values obtained from observations and experiments are $\mathbf{o}(\mathbf{x}) = (o(x_1), o(x_2), \dots, o(x_n))$.

‡ There are various definitions and debates about what constitutes a true discovery of scientific concepts and theories. In this article, we refer to discoveries in science that overturn old fundamental concepts and theories, rather than new findings within the existing theoretical framework.

In situations where multiple competing hypotheses $\mathcal{H} = h_1, h_2, \dots, h_l$ can explain certain features of a class of phenomena, but no single theory can explain all the samples simultaneously, our strategy is to identify the points of contradiction between these hypotheses and expand the theoretical space by not presupposing these points of contradiction. We refer to this approach as the ‘take back’ operation. Let $\mathbf{v} = (v_1, v_2, \dots, v_q)$ be the variables related to the points of contradiction. Instead of presupposing specific values for these variables, we take back and allow them to take arbitrary values within a reasonable range, thereby expanding the theoretical space. For each variable v_j , we consider a range of possible values $\mathcal{V}_j = [v_{j,min}, v_{j,max}]$, thus creating a larger theoretical space $\mathcal{H}^* = h_1^*, h_2^*, \dots, h_l^*$ (see Figure 1a). For each hypothesis h_i^* , the predicted values for the variables $\mathbf{x} = (x_1, x_2, \dots, x_n)$ are given by $\mathbf{p}_{h_i^*}(\mathbf{x})$.

2.1.2. Auto-Encoder Architecture for Aligning predictions of Theories with Observations

The second component of AI-Einstein Framework, using an auto-encoder which is a type of neural network commonly used in unsupervised learning and representation learning, to model the transformation from the predicted values of the scientific hypothesis to the actual values. The auto-encoder consists of an encoder g_ϕ and a decoder g_ψ . The encoder maps the input data to a lower-dimensional latent space representation, while the decoder reconstructs the original input from the latent representation.

In our specific design, instead of using the same input as the output, we use the predicted values $\mathbf{p}_{h_i^*}(\mathbf{x})$ as the input and the actual values $\mathbf{o}(\mathbf{x})$ as the output (see Figure 1b). The encoder maps the predicted values to the latent space representation $\mathbf{z} = g_\phi(\mathbf{p}_{h_i^*}(\mathbf{x}))$, and the decoder reconstructs the actual values from the latent representation, i.e., $\hat{\mathbf{o}}(\mathbf{x}) = g_\psi(\mathbf{z})$.

The objective of our auto-encoder is to minimize the reconstruction loss, which measures the difference between the actual values and their reconstructions:

$$\mathcal{L}(\phi, \psi) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{o}(\mathbf{x}_i) - \hat{\mathbf{o}}(\mathbf{x}_i)\|^2 \quad (1)$$

where N is the number of samples in the dataset.

2.1.3. Employing External Constraints for Theory Selection For each hypothesis h_i^* in the expanded theoretical space, we make predictions $\mathbf{p}_{h_i^*}(\mathbf{x})$ and train an auto-encoder to learn the transformation from these predictions to the actual observations $\mathbf{o}(\mathbf{x})$. we can extract the activation values $\mathbf{c} = (c_1, c_2, \dots, c_k)$ from the latent space representation $\mathbf{z}$. These activation values capture the essential information about how the predicted values of some hypothesis are transformed into the actual values. We can formalize the relationship between variables of different input samples and their corresponding activation values as a function $f_c : \mathbf{x}_i \rightarrow \mathbf{c}$ (see figure 1c). By analyzing the properties of this function, such as its monotonicity, linearity, or symmetry, we can determine which hypothesis best explains the observational data.

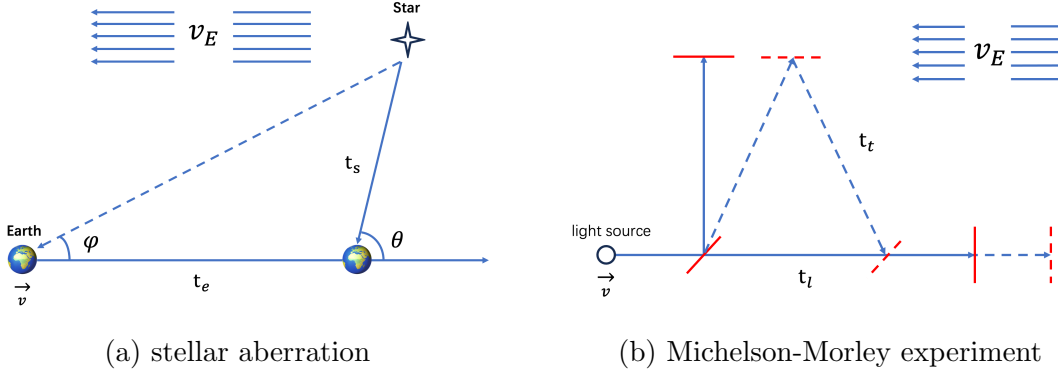


Figure 2: Schematic diagrams of the stellar aberration phenomenon and the Michelson-Morley experiment. (a) The phenomenon of stellar aberration refers to the variation in the apparent angle of distant stars (such as gamma Draconis) observed from Earth due to Earth’s annual motion, changing by up to plus or minus 20 arcseconds over the course of a year. (b) The Michelson-Morley experiment aimed to find the ether wind, that is, the velocity of ether relative to Earth.

For example, the hypothesis h_{best}^* that corresponds to the most symmetrical or consistent patterns in the relationship between $\mathbf{c}$ and the sample variables $\mathbf{x}_i$ can be considered the most plausible candidate for modification. The insights gained from analyzing the latent space of the auto-encoder associated with h_{best}^* can then be used to suggest how the hypothesis even background theory should be modified to better align with the observational data.

2.2. Application to a Historical Case Study

We apply our proposed framework to a classic case in the history of physics at the end of the 19th century—the theory of light propagation and measurements regarding ether, which led to the emergence of special relativity. This case is a typical scenario where multiple theories coexist, and although these theories have contradictions, they can explain some phenomena related to the propagation of light. However, no single theory could explain all related phenomena simultaneously.

We select two of the most classic related phenomena: the phenomenon of stellar aberration (Figure 2a) and the Michelson-Morley experiment (Figure 2b). Both phenomena and experiment are related to the propagation of light near the Earth’s surface. At the end of the 19th century, no single hypothesis could explain all related phenomena simultaneously. Table 1 lists the two most popular hypotheses at that time[22].

According to our method, in cases where mutually contradictory hypotheses can explain different samples of a phenomenon, we take a step back, avoiding presupposing the points of contradiction among competing hypotheses, and thereby expand the theoretical space. In this case, the point of contradiction is the velocity of ether. Thus,

Table 1: Hypotheses

Hypothesis	Description	Explicable phenomena	Point of contradiction
Fresnel’s ether drag	The ether exists and is stationary relative to the solar system	Stellar aberration	Velocity of ether relative to Earth equal to Earth’s orbital velocity
Stokes’ ether drag	The ether exists and can be completely dragged by Earth.	Michelson-Morley experiment	Velocity of ether relative to Earth is zero

when selecting the solar system as the stationary coordinate system, we expand the velocity of ether, v_E , to a range where $v_E \in [-50000, 50000]$ m/s, with each value representing a different theoretical assumption. We make predictions under different assumptions and compare them with actual observations and experiments.

In this case, both phenomena can be viewed as the encounter of two objects: the encounter of starlight and the observer on Earth in stellar aberration, and the encounter of two perpendicular light beams in the Michelson-Morley experiment. Therefore, we can use hypotheses from the expanded theoretical space to predict the time of the encounter of the two objects and use it as input, while the actually observed time serves as the output for supervision. For the phenomenon of stellar aberration, the input to the model is the predicted time t_e used by the observer on Earth under a specific hypothesis, while the output is the actual time t_s used by the starlight. Similarly, for the Michelson-Morley experiment, the input is the predicted time t_l under a hypothesis, and the output is the actual t_l (the sum of t_t and the experimentally measured dt).

Since historical data for stellar aberration and the Michelson-Morley experiment are indirect, we use historical data as a basis and make reasonable extensions to it. The details are provided in the appendix.

3. Results

We utilize the classic auto-encoder structure, setting the Earth’s orbital velocity v_l (with the solar system as the reference frame) as the sample variable, the predictions of hypotheses for two phenomena as inputs, and the actual observation as the output. We discovered that when the number of neurons in the intermediate encoding layer is reduced to one, the model can still accurately predict the output. The relationship between the activation value of the model’s intermediate layer **c** and the sample variable v_l is shown in Figure 3.

We can see that there is a clear quadratic change between the activation values

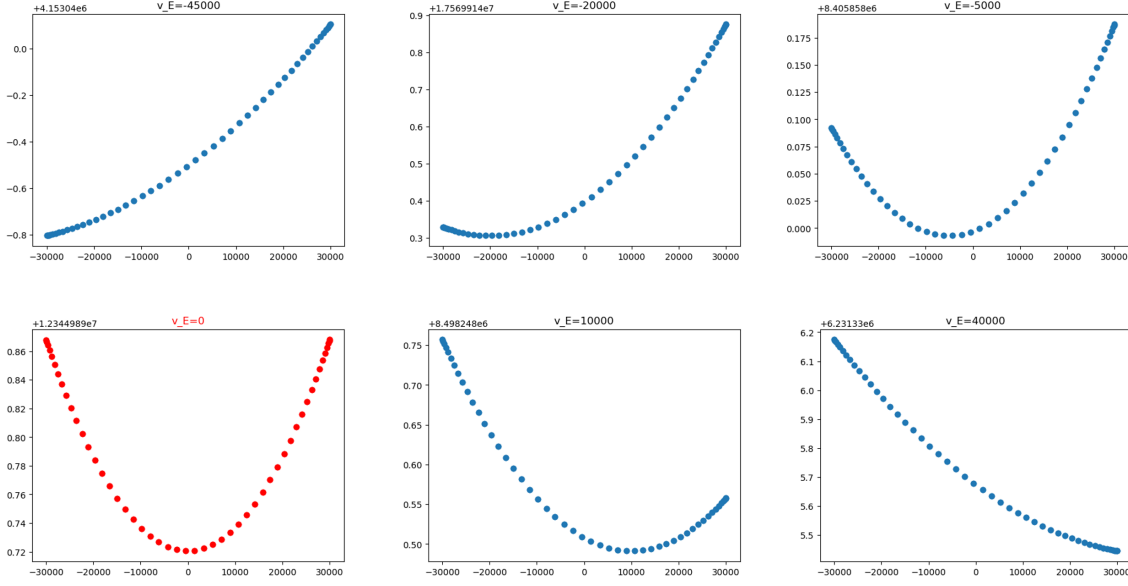


Figure 3: Relationship between the activation value of the model's intermediate layer $\mathbf{c}$ and the sample variable v_l (Earth's orbital velocity). The horizontal axis is v_l , and the vertical axis is $\mathbf{c}$, with v_l ranging from -30000m/s to 30000m/s.

of the middle layer and the Earth's orbital velocity. This indicates that, under these theories, the prediction for time(or space) needs to undergo a quadratic transformation to match the observational data, which contradicts the Galilean transformation assumed in physics at the end of the 19th century. According to the mainstream assumptions of that era, when the Earth's orbital velocity changes, the distance and time taken by light as input to the model should vary linearly.

When the velocity of ether relative to our designated "stationary" reference frame, $v_E = 0$, the graph is most symmetrical (over one Earth's annual motion cycle), which seems to be the most reasonable result. However, this does not prove that the $v_E = 0$ assumption is the best, because the setting of a stationary reference frame is arbitrary. Therefore, we relax the restrictions and attempt to view it from different reference frames, finding that the results are the same, as shown in Figure 4. This means that no matter from which reference frame we view it, the most symmetrical result occurs when the relative velocity of ether to Earth is 0. This also implies that the ether assumption might not be necessary, because according to Occam's razor, an object with a velocity of 0 relative to any reference frame can be considered as non-existent.

From the results, we received two hints on how theories should change: (1) Temporal and spatial parameters require a quadratic transformation (2) It might be feasible to assume there is no ether. These two points precisely match the two fundamental assumptions made by Einstein in his groundbreaking paper on special relativity in 1905[23]. We find that even without Einstein's genius, just based on historically known theories and a limited amount of indirect data, machine learning can identify these two

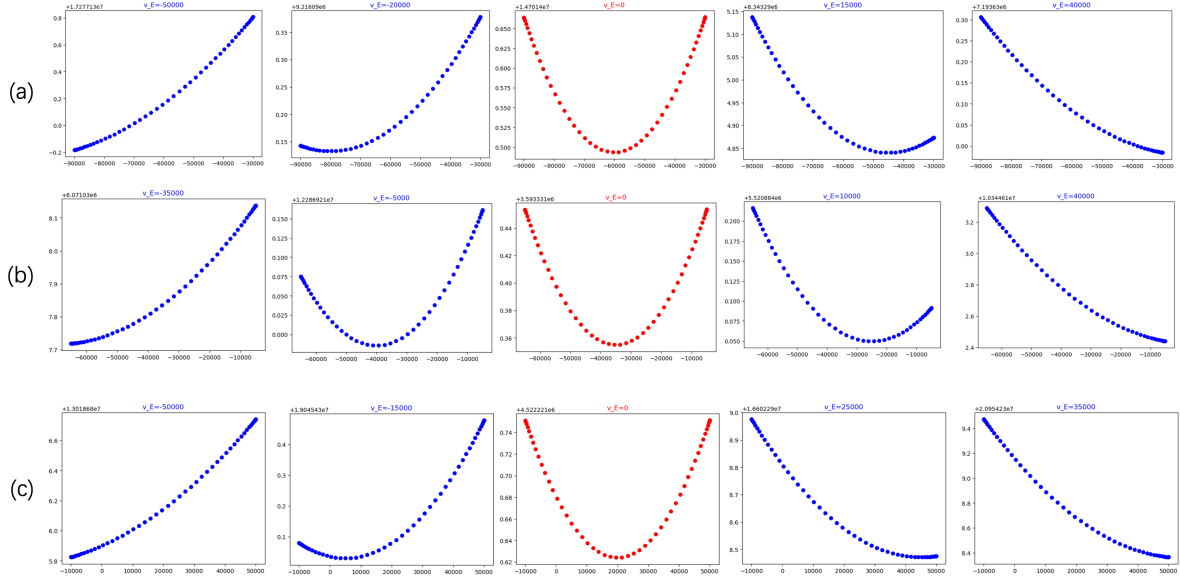


Figure 4: Relationship between the activation value of the model's intermediate layer $\mathbf{c}$ and the sample variable v_l (Earth's orbital velocity) under different reference frames. (a) v_l belongs to -90000m/s to -30000m/s . (b) v_l belongs to -60000m/s to -10000m/s . (c) v_l belongs to -10000m/s to 50000m/s

conceptual and ideological innovations.

4. Discussion

4.1. Conclusion

In this paper, we propose AI-Einstein framework, which showcases the potential of machine learning to suggest new scientific concepts and theories, even with limited and indirect historical data. By applying this framework to the late 19th-century debate on light propagation and ether, we've aligned with Einstein's revolutionary suggestions on special relativity without relying on direct data. This approach not only validates the feasibility of AI-driven scientific re-discovery but also emphasizes the necessity of integrating machine learning with scientific philosophy and historical analysis to unlock novel scientific insights.

4.2. Future work

In this paper, we introduced the AI-Einstein framework, which significantly relies on human input for defining and refining several critical aspects of the scientific discovery process. Specifically, the first and third components of our framework still require manual intervention. Identifying the points of contradiction among competing scientific hypotheses and expanding the theoretical space, as well as filtering hypotheses based on criteria such as symmetry and simplicity, are tasks that cannot be entirely formalized.

These processes inherently depend on human insight, particularly in leveraging common-sense knowledge that lies outside strict scientific theories.

Looking forward, we aim to address these challenges by harnessing the capabilities of generative large language models (LLMs). These advanced models offer potential in partially replacing the need for human intuition in the scientific process, especially in identifying contradictions and expanding theoretical frameworks without explicit human direction. By automating these aspects, we hope to reduce the subjective bias in hypothesis selection and enhance the efficiency and objectivity of scientific discovery.

Additionally, we plan to continue searching for more real-world examples to further validate and refine our proposed framework. By applying AI-Einstein to a broader range of historical and contemporary scientific problems, we aim to demonstrate its versatility and effectiveness in uncovering new scientific insights across various domains. This continued application is essential for iteratively improving the framework and adapting it to accommodate the complex nature of real-world scientific inquiry.

5. Appendix

5.1. Historical Data Processing

This paper uses two sets of historical data can be see in Figure 5: stellar aberration observation data and Michelson-Morley experiment data.

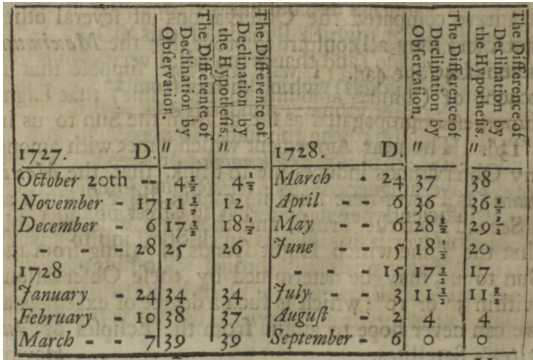


Figure 5(a) is a historical manuscript page showing Bradley's observations of the aberration of light for γ Draconis in 1728. The page is divided into two main columns for the years 1727 and 1728. Each column has sub-columns for 'The Difference of Declination by the Hypothesis' and 'The Difference of Declination by Observation'. The data is organized by month, with entries for October, November, December, January, February, March, April, May, June, July, August, and September. The values are given in degrees and minutes, with some entries in parentheses indicating specific observations or calculations.

(a) Bradley's observation data in 1728

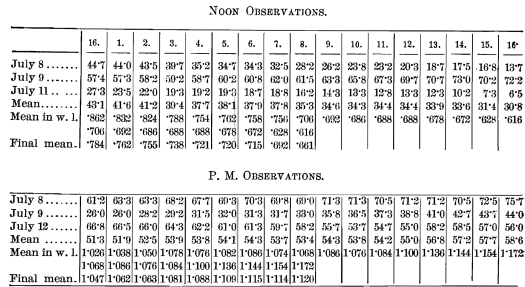


Figure 5(b) is a table showing the Michelson-Morley experiment data in 1872. The table is divided into two main sections: 'NOON OBSERVATIONS' and 'P. M. OBSERVATIONS'. Each section has a header row with numbers 1 through 16, representing different sets of data. The data is organized by date, with entries for July 8, July 9, July 11, July 12, and July 13. The values are given in degrees and minutes, with some entries in parentheses indicating specific observations or calculations. The table also includes a 'Final mean' row for each section.

(b) Michelson-Morley experiment data in 1872

Figure 5: Historical data (a) Bradley's observation of the aberration of light for γ Draconis in 1728, including the deviation observed by Bradley and the deviation he calculated based on the corpuscular theory of light. (b) Michelson-Morley experiment data in 1872, including 16 sets of data. They represent the readings of the micrometer screw gauge, which measures the shifts in the interference fringes observed through their experimental setup.

For the stellar aberration observation data, we chose Bradley's 1728 observation data[24], which consists of the angular deviation of γ Draconis over a year (Figure 5a). We used these observational data, along with the actual angle between γ -Draconis

and the ecliptic($51^\circ 29' 20''$), to infer the angle between the star and the ecliptic at the time of Bradley's observation. We obtained the variation(ab_B) in the angle between the star and the ecliptic that Bradley could observe in his era over a year. According to Bradley's hypothesis at the time, which treated light as particles, the predicted stellar aberration required the Earth's orbital velocity. Therefore, based on Bradley's predicted data(ab_{hyp}), we inferred the Earth's orbital velocity he used during his observations (v_e), see Table 2. We ultimately obtained 16 sets of data, which are the Earth's orbital velocities and their corresponding observed angles between the star and the ecliptic. We then used interpolation to expand the data to 100 sets (using 100 equally spaced Earth's annual orbital velocities). Finally, we obtained the velocity v_{ex} of the Earth in 100 time points during its annual motion relative to the solar reference system and the corresponding observed angle ϕ_{ex} between the star and the ecliptic at each moment.

Table 2: Bradley's real observation data in 1728 and processing data

ab_B	ϕ	ab_{hyp}	v_e
0	1.308301231363355	0	29347.25419213127
4.5	1.3083230479790051	4.5	22574.678602963657
11.5	1.3083569849366827	12	11287.229049502861
17.5	1.3083860737575492	18.5	1504.9511343564739
25	1.3084224347836324	26	-9782.08684105682
34	1.3084660680149323	34	-21821.351046015407
38	1.3084854605621767	37	-26336.01047061413
39	1.3084903086989879	39	-29345.763830594897
37	1.3084806124253656	38	-27840.88910950535
36	1.3084757642885545	36.5	-25583.569681953366
28.5	1.3084394032624713	29.5	-15049.295789218235
18.5	1.3083909218943603	20	-752.4740973889411
17.5	1.3083860737575492	17	3762.385185004081
11.5	1.3083569849366827	11.5	12039.71882588688
4	1.3083206239105996	4	23327.183080901836
0	1.308301231363355	0	29347.25419213127

For the Michelson-Morley experiment data, we used the 1872 experimental data[25], which consisted of 16 sets (Figure 5b). The data in the figure represent the time difference between the convergence of two light beams (obtained through light interference fringes). The instrument was rotated one revolution, and measurements were taken 16 times to obtain 16 sets of data. We similarly used interpolation to expand the data to 100 sets.

5.2. Unified Design of Two Phenomena

Both stellar aberration and the Michelson-Morley experiment involve the propagation of light near the Earth's surface. Although both are related to light and the properties of the ether believed at the time, in real scientific research scenarios, it is generally impossible to obtain the desired direct data, such as the space-time data of light. Instead, indirect data are obtained, such as the deviation angle in stellar aberration and the interference effect of two light beams in the Michelson-Morley experiment. Therefore, if we want to find a theory that can simultaneously explain the underlying principles of these two phenomena, we need to unify the data of these two phenomena into the same type of phenomenon so that they can be used together as input. Both stellar aberration and the Michelson-Morley experiment can be viewed as the encounter of two moving objects (light and the observer). Therefore, we can use the prediction of a certain variable in the encounter event under a specific hypothesis as the model's input, while the observed value of this variable serves as the output. For stellar aberration, as shown in Figure 2a, we use the observer's motion time t_e in a specific coordinate system (such as the solar coordinate system) to predict the trajectory time of γ Draconis starlight which is t_{s-pre} . We let t_{s-pre} as the input and real t_s which is equal to t_e as the output. Let c be the speed of light, v be the component of the Earth's orbital velocity in the observation direction, θ be the true angle of the star, and ϕ be the observed angle (with stellar aberration), v_E be the velocity of the ether relative to the reference we choice. Then, the travel time of starlight:

$$t_{s-pre} = \frac{vt_e}{\frac{\sin \theta}{\tan \phi} - \cos \theta} / c_1 \quad (2)$$

$$c_1 = \sqrt{c^2 + v_E^2 - 2 \cos \theta \cdot c \cdot v_E} \quad (3)$$

For the Michelson-Morley experiment, as shown in Figure 2b, predicted value of t_l under a certain theoretical assumption as the input, and the actual value t_l (the sum of t_t and the experimentally measured dt) as the output. Let t_{l-pre} be predicted value, L be the length of the light path, v be the component of the Earth's orbital velocity in the experimental direction relative to the solar reference system, and dt be the time difference between the two light beams. Then, we have:

$$t_{l-pre} = \frac{L}{c + v_E - v} + \frac{L}{c - v_E + v} \quad (4)$$

$$t_l = t_t + dt = \frac{2L}{\sqrt{c^2 + v_E^2 - v^2}} + dt \quad (5)$$

In simpler terms, the input of the model is the predicted value under different theoretical assumptions, and the output is the actual observed value or the value that can be deduced from the actual observed value under this theory.

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